

42.510 Statistical Learning for Data Science

Matrix factorization I: SVD applied to image compression

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Week 9, Lecture 1

Outline

- Unsupervised learning vs. Supervised learning
- Image Compression
- Matrix factorization
 - ▶ Eigendecomposition (leading to PCA)
 - ▶ Singular Value Decomposition (SVD)
- R implementation

Supervised vs unsupervised learning

	Supervised learning	Unsupervised learning
Data	Data is labeled. Output: y , input: $x_1, \dots, x_p$.	Unlabeled data. Only input: $x_1, \dots, x_p$.
Goal	Predicting the response, classifying, etc.	Not direct. Understanding the structure. Clustering, dimension reduction
Assessment	Break into training and test sets and validate.	Difficult.
Methods	Regression (linear, logistic, ...), Random forests, decision trees, ...	k-means, hierarchical clustering, PCA, SVD, ...

What's the difference?



What's the difference?



Image sharing

- Image sharing: Whatsapp, Snapchat, Facebook, Instagram, etc.
- ~ 14 billion per day (2024). Used to be ~ 1.8 billion in 2014.
- Images are not shared at best quality. But, compression needs to maintain
 - ▶ minimum quality
 - ▶ user experience

Image storage and compression

- A digital image: finite number of pixels arranged as a matrix.
- Grayscale images contain a value representing the amount of light from 0 to 1 at each pixel.
- Color images may have a set of stacked colors depending on format: RGB, CMYK
- Compression can be *lossy* or *lossless*. Here we look at a *lossy* compression.
- Use matrix factorization for data compression
 - ▶ Singular value decomposition

Eigendecomposition or Spectral decomposition

- Consider an $n \times n$ matrix $\mathbf{A}$ with real-valued entries.
 - $\lambda \in \mathbb{R}$ is an **eigenvalue** with corresponding **eigenvector** $\mathbf{u} \in \mathbb{R}^n$ if

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}.$$

- Theorem:** Consider a *symmetric* $n \times n$ matrix $\mathbf{A}$ with real-valued entries. Then $\mathbf{A}$ has n orthonormal eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ such that

- a) $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j, \quad j = 1, \dots, n,$
- b) $\mathbf{u}_j^\top \mathbf{u}_j = 1; \quad i = 1, \dots, n, \text{ or } \|\mathbf{u}_j\| = 1,$
- c) $\mathbf{u}_i^\top \mathbf{u}_j = 0; \quad i \neq j, i, j \in \{1, \dots, n\}.$

- Customarily we write the eigenvalues in decreasing order:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.$$

- We can now write $\mathbf{A} = \sum_{j=1}^n \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$.
- The number of non-zero eigenvalues of $\mathbf{A} = \text{rank}(\mathbf{A})$.

Eigendecomposition or Spectral decomposition (cont'd)

- $\mathbf{A} \in \mathbb{R}^{n \times n}$ is *positive semidefinite* (PSD), also sometimes called *non-negative definite* (NND), if

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

- **Example:** The sample Variance-Covariance matrix given by $\mathbf{S}$ or $\hat{\Sigma}$ is PSD.
- For PSD (or NND) matrices, we have all the eigenvalues to be non-zero and hence we can write them as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0.$$

- In fact, if $\text{rank}(\mathbf{A}) = r$, then

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > \lambda_{r+1} = \dots = \lambda_n = 0,$$

and we have $\mathbf{A} = \sum_{j=1}^r \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$.

Matrix view (eigendecomposition)

- We have

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$$

where

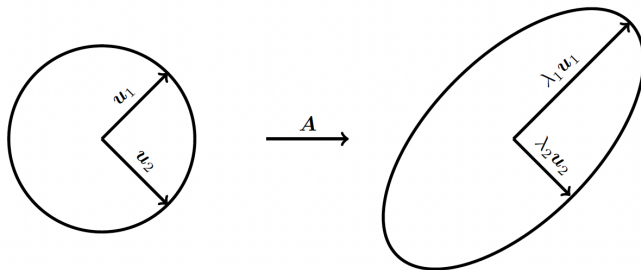
$$\begin{bmatrix} \mathbf{A} \end{bmatrix} = \underbrace{\begin{bmatrix} | \\ \mathbf{u}_j \\ | \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} - & \mathbf{u}'_j & - \end{bmatrix}}_{\mathbf{U}^\top}$$

- If $\text{rank}(\mathbf{A}) = r \leq n$ then

$$\lambda_1 \geq \dots \lambda_r > \lambda_{r+1} = 0 = \dots = \lambda_n = 0.$$

- Also note $\mathbf{U}^\top \mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}$.

Geometric view (eigendecomposition)



Optimization view (eigendecomposition)

Claim 1:

$$\lambda_1 = \max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \mathbf{A} \mathbf{u}$$

where λ_1 is the maximum eigenvalue with corresponding eigenvector $\mathbf{u}_1$ being the maximizer.

Optimization view (eigendecomposition)

Claim 2:

$$\lambda_2 = \max_{\|\mathbf{u}\|=1, \mathbf{u} \perp \mathbf{u}_1} \mathbf{u}^\top \mathbf{A} \mathbf{u}$$

where λ_2 is the maximum eigenvalue with corresponding eigenvector $\mathbf{u}_2$ being the maximizer satisfying both $\|\mathbf{u}_2\| = 1$ and $\mathbf{u}_2 \perp \mathbf{u}_1$.

Useful fact

$$\text{trace}(\mathbf{A}) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i = \text{trace}(\mathbf{\Lambda}).$$

Singular value decomposition

- Consider an $m \times n$ matrix $\mathbf{A}$ with real-valued entries.
- Let $\text{rank}(\mathbf{A}) = r \leq \min(m, n)$.
- We can write

$$\mathbf{A} = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top$$

for $\sigma_1 \geq \dots \geq \sigma_r > 0$.

- We call
 - ▶ $u_1, \dots, u_r$ the **left** singular vectors (in $\mathbb{R}^m$),
 - ▶ and, $v_1, \dots, v_r$ the **right** singular vectors (in $\mathbb{R}^n$).
- Also note that **left** singular vectors are orthonormal, and so are the **right** singular vectors:
 - ▶ $u_j^\top u_j = 1, v_j^\top v_j = 1, \forall j = 1, \dots, r;$
 - ▶ $u_i^\top u_j = 0, v_i^\top v_j = 0, \forall i \neq j \in \{1, \dots, r\}.$

Matrix view (SVD)

- We have

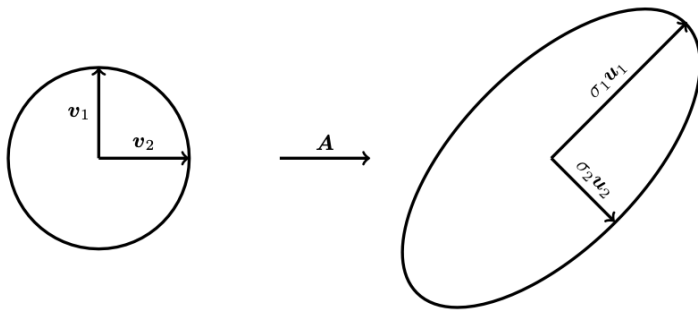
$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

where

$$\underbrace{\begin{bmatrix} \mathbf{A} \end{bmatrix}}_{m \times n} = \underbrace{\begin{bmatrix} \mathbf{U} \end{bmatrix}}_{m \times r} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}}_{\substack{\mathbf{\Sigma} \text{ is } r \times r \text{ matrix}}} \underbrace{\begin{bmatrix} \mathbf{V}^\top \end{bmatrix}}_{r \times n}$$

- Here $\mathbf{U} = [\mathbf{u}_1 \dots \mathbf{u}_r]$ and $\mathbf{V} = [\mathbf{v}_1 \dots \mathbf{v}_r]$.
- Both $\mathbf{U}$ and $\mathbf{V}$ are **semi-unitary**, i.e., $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}$.
- Note: $\mathbf{U}$ and $\mathbf{V}$ are not necessarily square matrices.

Geometric view (SVD)



Relationship between eigendecomposition and SVD

- Suppose $\mathbf{A} \in \mathbb{R}^{m \times n}$. Let $\text{rank}(\mathbf{A}) = r$ and the SVD is $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$.
- Then we have

$$\mathbf{A}\mathbf{A}^\top = \mathbf{U} \begin{bmatrix} \sigma_1^2 & 0 & \dots & \dots \\ 0 & \sigma_2^2 & 0 & \dots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 & \sigma_r^2 \end{bmatrix} \mathbf{U}^\top.$$

- Columns of $\mathbf{U}$ are eigenvectors of $\mathbf{A}\mathbf{A}^\top$ with corresponding eigenvalues $\sigma_1^2, \dots, \sigma_r^2$.
- Similarly considering

$$\mathbf{A}^\top \mathbf{A} = \mathbf{V} \begin{bmatrix} \sigma_1^2 & 0 & \dots & \dots \\ 0 & \sigma_2^2 & 0 & \dots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 & \sigma_r^2 \end{bmatrix} \mathbf{V}^\top,$$

we have columns of $\mathbf{V}$ are eigenvectors of $\mathbf{A}^\top \mathbf{A}$ with corresponding eigenvalues $\sigma_1^2, \dots, \sigma_r^2$.

Optimization view (SVD)

- **Claim:**

$$\sigma_1 = \max_{\|\mathbf{v}\|=1} \|\mathbf{A}\mathbf{v}\|$$

where σ_1 is the maximum singular value with corresponding right singular vector $\mathbf{v}_1$ being the maximizer.

Low-rank approximation with SVD

- We know

$$\mathbf{A} = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top.$$

- For a low-rank approximation we drop the lower singular values to approximate $\mathbf{A}$ by

$$\hat{\mathbf{A}} = \sum_{j=1}^k \sigma_j \mathbf{u}_j \mathbf{v}_j^\top$$

where $k < r$.

- Here we only use the k largest singular values of $\mathbf{A}$, i.e., $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$.

Why is a low-rank approximation useful?

① Reduction in storage space

- ▶ Suppose $m = n = 1000$ and $k = 20$.
- ▶ Storage for an $m \times n$ matrix:

$$m \times n = 1000 \times 1000 = 1,000,000.$$

- ▶ Storage for its rank k approximation:

$$m \times k + n \times k + k = 1000 \times 20 + 1000 \times 20 + 20 \sim 40,000.$$

② In some cases, a low rank approximation is perhaps a natural representation of reality.

Quality of a low-rank approximation

- Measuring the quality of approximation: $\|\mathbf{A} - \hat{\mathbf{A}}\|$.
- Use Frobenius norm:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}.$$

- **Claim:** $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}$.

R implementation

- We work on a grayscale image (and a color image).
- Feel free to use your own jpeg image.

References

- Teaching notes